

Solon Falas

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Professional Summary

Machine Learning Engineer / Data Scientist with a PhD in Computer Engineering, 7+ years applied research experience, and ML-focused work since 2020–2021. Designed and deployed physics-informed neural networks for real-world cyber-physical systems across the full ML lifecycle — data preprocessing, feature engineering, model training (TensorFlow, PyTorch), hyperparameter optimisation (Optuna), and reproducible evaluation.

Technical Skills

Programming: Python, Bash/Shell, C/C++/C#, L^AT_EX

ML / Deep Learning: TensorFlow, PyTorch, scikit-learn, Optuna

Data Science: pandas, NumPy, Matplotlib, Seaborn, Jupyter

Tools: Linux, Git, MATLAB, PowerWorld

AI Tooling: Claude, Codex, GitHub Copilot, LLM-assisted R&D, agentic workflows, spec/test-driven development, prompt engineering

Domains: ML model development, data pipelines, power systems, IoT security

Education

Ph.D., Computer Engineering (2017–2025)

University of Cyprus, Nicosia, Cyprus

Cross-Layer Physics-based Security for Cyber-Physical Systems

B.Sc., Computer Engineering (2013–2017)

University of Cyprus, Nicosia, Cyprus

Awards

- **Best Computer Engineering Undergraduate Dissertation**, University of Cyprus (2017)
- **Ph.D. Scholarship**, KIOS Research and Innovation Center of Excellence (2017–2025)

Hobbies & Activities

- Personal finance, portfolio construction, risk analysis, investing
- Running, weightlifting, fitness, sports
- Digital privacy, DIY self-hosting, smart-home projects

Languages

Greek (Native) | English (Fluent)

Work Experience

Research Associate – Machine Learning & Security (2025–Present)

KIOS Research and Innovation Center of Excellence, University of Cyprus

- Conducting applied ML research and engineering on cyber-physical systems and embedded security.

Researcher – Machine Learning & Data Science (2017–2025)

KIOS Research and Innovation Center of Excellence, University of Cyprus

- Built and evaluated physics-informed neural networks (PINNs) for power-system state estimation under noisy, faulty, and adversarial operating conditions, embedding AC power-flow constraints as physics-based regularisation to improve generalisation.
- Built end-to-end Python data pipelines to ingest, preprocess, and curate multi-scenario simulation datasets; ran systematic hyperparameter optimisation with Optuna and analysed estimation-error metrics across diverse operating conditions.
- Maintained reproducible ML workflows with version-controlled code, documented configurations, and benchmarking reports using Python, TensorFlow, PyTorch, and MATLAB.

Teaching Assistant – Computer Engineering (2017–2025)
University of Cyprus

- Delivered lab instruction and technical support for digital logic, microprocessors, operating systems, and VLSI CAD courses.

Selected ML Projects

Robust Power-System State Estimation with Physics-Informed Neural Networks

Built TensorFlow/PyTorch PINN models with AC power-flow regularisation; ran Optuna hyperparameter sweeps and benchmark evaluation across fault and adversarial-measurement scenarios; established reproducible training and reporting workflow.

Learning Without Adversarial Training: A PINN for Secure Power System State Estimation under False Data Injection Attacks

Developed a physics-informed neural network that stays robust to false data injection attacks without adversarial training, using embedded power-flow constraints to resist manipulated measurements while preserving accuracy under normal operating conditions.

Selected Publications

- S. Falas et al., “Robust Power System State Estimation Using Physics-Informed Neural Networks,” *IEEE Transactions on Industrial Informatics*, 2025.
- S. Falas et al., “Data Manipulation Attack Mitigation in Power Systems Using Physics-Informed Neural Networks,” *IEEE CSR*, 2025.
- S. Falas et al., “A Modular End-to-End Framework for Secure Firmware Updates on Embedded Systems,” *ACM JETC*, 2022.